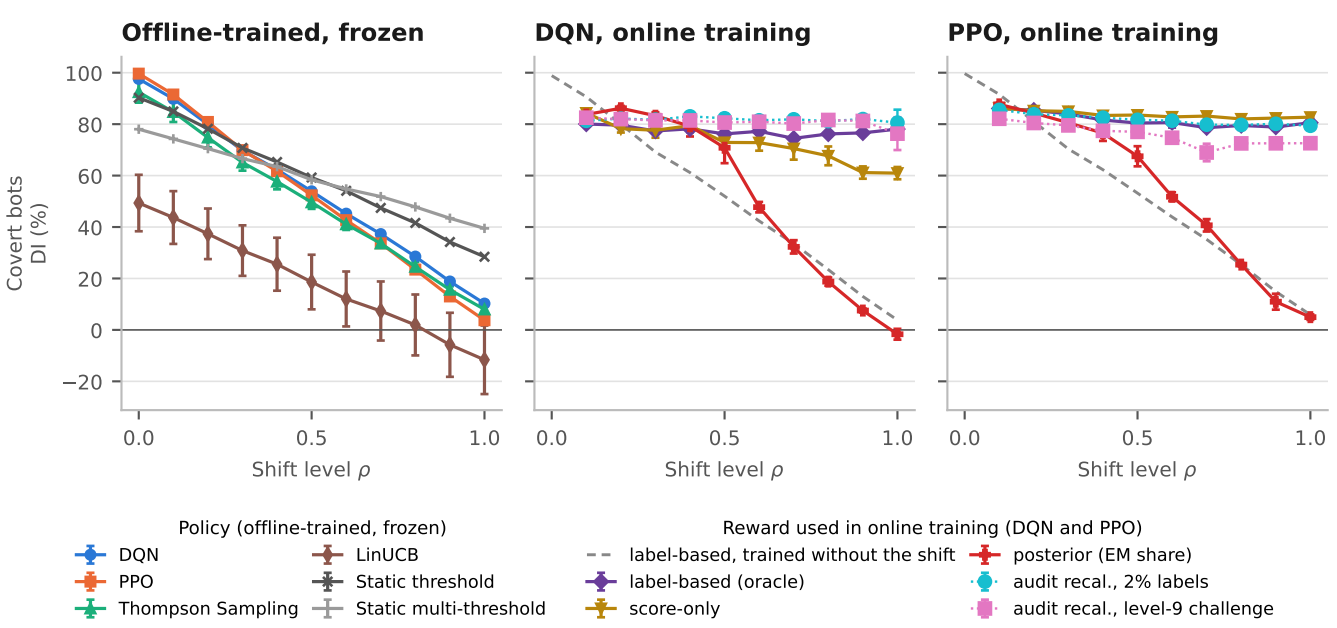
Offline-trained, frozen
DQN, online training
PPO, online training
Covert bots DI (%)
100
80
60
40
20
0
−20
0.0
0.5
1.0
Shift level ρ
Policy (offline-trained, frozen)
DQN
PPO
Thompson Sampling
LinUCB
Static threshold
Static multi-threshold
Reward used in online training (DQN and PPO)
label-based, trained without the shift
label-based (oracle)
score-only
posterior (EM share)
audit recal., 2% labels
audit recal., level-9 challenge